THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** S^{2-}	Z = 16 is sulphur, 2 extra electrons.
3	**(c)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 20 and 16	X lost 2, Y gained 2.
6	**(a)** 20	10x + 11(100 - x) = 1080.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(d)** argon	18 electrons.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Alpha particles pass through the empty space and are deflected only when they come close to the tiny nucleus.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the discovery of the neutron is not the reason for the model's failure (a revolving electron should lose energy and spiral into the nucleus).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1).	1+1
12	Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1).	1+1
13	No dense positive centre in his model (1); It predicted only small deflections (1).	1+1
14	Elements are mixtures of isotopes (1); Atomic mass is the weighted average (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	2, 8, 3 (1); M^{3+} (1); Mass = 54 + 48 = 102 u (1).	1+1+1
16	p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1).	1+1+1
17	Electrons occupy fixed shells (1); No radiation in these states (1); Energy changes only by jumps (1).	1+1+1
	OR: Same valence electrons in a group (1); Across a period valence electrons increase 1 to 8 (1); Chemical behaviour depends on valence electrons (1).	1+1+1
18	Splits on neutron capture giving energy (1); Chain reaction controlled in reactors (1); Radioactive waste and accidents (1).	1+1+1
	OR: Atom had 20 electrons (1); Z = 20 (1); Calcium (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Atomic number = number of protons = 11; mass number = 11 + 12 = 23 (1). (ii) 2, 8, 1 (1). (iii) Valency = 1 (it has 1 valence electron) (1). (iv) The atom loses its one outer electron (1) to give X^{+} with 10 electrons and configuration 2, 8 (1).	1+1+1+2
20	Closest approach of alpha particles gives an upper limit for the nucleus size (1); Atom radius about 10^-10 m (1); Density = mass/volume is huge (1); About 10^18 kg per cubic metre (1); Nucleus holds nearly all the mass (1).	1+1+1+1+1
	OR: Li, Na, K have one valence electron (1); Form +1 ions (1); F, Cl, Br have seven valence electrons (1); Form -1 ions (1); Same group, similar properties (1).	1+1+1+1+1

Mistakes to avoid:

- Confusing atomic number with mass number, or finding neutrons as $A + Z$ instead of $A - Z$.

- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.
- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	Z = 15: 2, 8, 5.
2	**(c)** 207	Sum.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** Z^{2-}	Gain 2 to make 8.
5	**(a)** tritium	2 : 1.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** hydrogen-1	0 neutrons.
8	**(d)** $^{14}_{6}C$	14 - 6 = 8.
9	**(c)** A is true but R is false.	A is true; electrons are about 1/1836 of the mass of a proton, so R is false.
10	**(d)** A is false but R is true.	35.5 u is the average atomic mass; individual atoms have mass number 35 or 37, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Z = 9: fluorine (1); $^{19}_{9}F^{-}$ (1).	1+1
12	16 + 4 x 8 = 48 (1); Add 2 for the charge: 50 (1).	1+1
13	Charge +3 (1); 2, 8: resembles neon (1).	1+1
14	Neutrons are uncharged (1); Alpha particles are repelled by the positive nucleus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(1810 + 6.3 + 202.4)/100 (1); = 20.19 u (1); About 20.2 u (1).	1+1+1
16	Atom: neutral $^{23}_{11}Na$ (1); Ion: Na^{+} has lost an electron (1); Isotope: $^{24}_{11}Na$ has a different number of neutrons (1).	1+1+1
17	Mg (2,8,2) loses 2 electrons (1); S (2,8,6) gains 2 (1); Both reach the octet (1).	1+1+1
	OR: Atom: 11, 12, 11 (1); Ion: 11, 12, 10 (1); Ion is smaller and stable (1).	1+1+1
18	Isobars (1); $^{14}_{6}C$ and $^{14}_{7}N$ (1); Different chemical properties (1).	1+1+1
	OR: (10 x 1 + 11 x 4)/5 (1); = 10.8 u (1); Boron (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Atoms of the same element with the same atomic number but different mass numbers (1). (ii) Boron-10: $10-5=5$ neutrons; boron-11: $11-5=6$ neutrons (1). (iii) $0.20\times10+0.80\times11=2+8.8=10.8$ u (1). (iv) They have the same number of electrons and the same electronic configuration (1); use: cobalt-60 for cancer treatment, iodine-131 for thyroid, carbon-14 dating (any one) (1).	1+1+1+2
20	Li 2,1 valency 1 (metal); Be 2,2 valency 2 (1); B 2,3 valency 3 (1); C 2,4 valency 4 (1); N 3, O 2, F 1 (1); Ne 0; trend 1 to 4 and back to 0 (1).	1+1+1+1+1
	OR: H: 1, 0, 1 (1); He: 2, 2, 2; C: 6, 6, 6 (1); O2-: 8, 8, 10 (1); Na+: 11, 12, 10 (1); Cl-: 17, 18, 18 (1).	1+1+1+1+1

Mistakes to avoid:

- Confusing atomic number with mass number, or finding neutrons as $A+Z$ instead of $A-Z$.
- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8-5=3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.

- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** They have different numbers of electrons	Electron numbers are equal.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** 1 km	Ratio 10^5 cm = 1 km.
4	**(c)** 10	11 - 1.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 2, 7, 1	L shell must be full before M.
7	**(a)** neon	Mg2+: 2, 8.
8	**(c)** A is true but R is false.	A is true; R is false: neutrality is due to equal protons and electrons.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A filled octet gives a stable configuration with valency zero.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the chemical properties are the same because the number of electrons is the same, not because of the mass numbers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass is 1/1836 u, negligible (1); Its charge balances the protons (1).	1+1
12	Same number of valence electrons (1) (1); Li 2,1; Na 2,8,1; K 2,8,8,1 (1).	1+1
13	H-1: 1 nucleon (1); D: 2 nucleons; T: 3 nucleons (1).	1+1
14	Z = 17 (1); 7 valence electrons (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Alpha particles are helium nuclei (+2) (1); Repelled by the positive nucleus (1); Path bends away (1).	1+1+1
16	Atomic number is unique to each element (1); Isotopes have the same Z but different A (1); Isobars have the same A but different Z (1).	1+1+1
17	Tritium has two extra neutrons (1); Same electron arrangement (1); Chemical properties depend on electrons (1).	1+1+1
	OR: Both have 2, 8, 8 (1); Argon is neutral with a full shell (1); K^+ has a stable octet but is charged and exists in compounds (1).	1+1+1
18	P 2,8,5: 3 (1); S 2,8,6: 2; Cl 2,8,7: 1 (1); Ar 2,8,8: 0 (1).	1+1+1
	OR: Alpha particles are heavy and charged (1); They ionise and lose energy quickly (1); Gamma rays are uncharged and penetrating (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Ernest Rutherford (with Geiger and Marsden) (1). (ii) Most of the atom is empty space (1). (iii) The centre is positively charged, since positive alpha particles were repelled (1). (iv) Any two: the nucleus is very small compared with the atom; it is very dense and holds nearly all the mass; it is positively charged; electrons move around it (1+1).	1+1+1+2
20	Electron: Thomson, plum pudding (1); Proton: Rutherford, nucleus (1); Neutron: Chadwick, mass puzzle (1); Bohr: energy shells (1); Models changed with evidence (1).	1+1+1+1+1
	OR: Complete outer shell (1); Very weak forces between atoms (1); Low boiling points (1); Uses: helium in balloons, neon in lights, argon in bulbs (1); Unreactive nature (1).	1+1+1+1+1

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